

Diagnosing Transformation-Order and Inverse-Domain Errors

Telling inside changes from outside changes, ordering transformations correctly, and choosing the right inverse branch

Directions

- Each problem has two jobs: compute the correct value, and say in words what went wrong. Both are graded.
- A diagnosis names the rule that was broken. “They got a different number” is a check, not a diagnosis.
- Show the substitution you used — one well-chosen input settles most of these arguments.

1. Let $f(x) = x^2$. A student claims that $g(x) = f(x) + 5$ and $h(x) = f(x + 5)$ have the same graph.

(a) Compute $g(2)$ _____ and $h(2)$ _____

(b) Name the error in the student’s claim, saying what each 5 actually does.

Answer: _____

2. To graph $y = 2(x - 3)^2 + 1$ starting from $y = x^2$, a student shifted up 1 first, then stretched vertically by 2, then shifted right 3, and reported a vertex height of 2.

- (a) Evaluate the printed rule $y = 2(x - 3)^2 + 1$ at $x = 3$ to find the true vertex height.

Answer: _____

- (b) Name the error in the student's order of operations.

Answer: _____

3. A student claims that the inverse of $f(x) = x^2$ is $f^{-1}(x) = \sqrt{x}$, with no restriction placed on f .

- (a) Compute $f(-4)$ _____ and then the square root of that result _____

- (b) Say what restriction f must carry before $\sqrt{x}$ can be its inverse.

Answer: _____

4. Maya was asked for the rule that stretches $y = |x|$ vertically by 3 and then shifts it down 4. She wrote $g(x) = 3(|x| - 4)$ and reported $g(2) = -6$.

(a) Write the correct rule and use it to find the correct $g(2)$.

Answer: _____

(b) Name her error, saying which number was multiplied when it should not have been.

Answer: _____

5. Let $f(x) = \sqrt{x - 2}$. A student wrote $f^{-1}(x) = x^2 + 2$ and said its domain is all real numbers.

(a) Using the student's rule, compute $f(f^{-1}(-3))$.

Answer: _____

(b) Explain why that result shows the domain claim is wrong, and give the restriction f^{-1} needs.

Answer: _____

6. Let $f(x) = \sqrt{x}$. A student says the graph of $y = f(2x)$ is the graph of f stretched horizontally by a factor of 2.

(a) Solve $f(2x) = 4$ _____ and solve $f(x) = 4$ _____

(b) Say whether the factor 2 stretches or compresses the graph horizontally, using your two answers as the evidence.

Answer: _____

7. A student described $y = -(x + 1)^2 + 5$ as: reflect $y = x^2$ in the x -axis, shift left 1, shift up 5 — but carried out the reflection last, and reported a vertex height of -5 .

(a) Evaluate the printed rule at $x = -1$ to find the true vertex height.

Answer: _____

(b) Name the error, saying which step must come before which.

Answer: _____

8. Let $f(x) = x^2 - 4x + 7$ with the domain restricted to $x \leq 2$. A student wrote $f^{-1}(x) = 2 + \sqrt{x - 3}$ and so reported $f^{-1}(12) = 5$.

(a) Complete the square, then find the correct value of $f^{-1}(12)$.

Answer: _____

(b) Explain which branch of the parabola the restriction keeps, and why that fixes the sign of the square root.

Answer: _____